WEEK 3 WARM-UP EXERCISES

JAVA SYNTAX DETAILS

1. The keyword **void** before the name of the method means that the method does not return anything.
2. Create a method that returns an integer as follows:

```
private static int myMethod() {
    return 42;
}
```

3. The System.out.**print** method prints output **without a line break**.
4. The following gives you the number of elements in the array: `myArray.length`
5. The following returns the length of the string: *firstname*.**length()**

→ To insert `System.out.println()` quickly, type **syso**, then press **Ctrl+Space**

→ To format your code correctly, press **Ctrl+Shift+F**

WARM-UP EXERCISES (Not to be submitted !)

Create these programs in the ***warmup*** package in the ***Programming_1*** project.

In this set of exercises, you are not required to handle any empty arrays.

1. Write a program named **WarmUp_10** that includes a method named **printArray**. This method should print the values in a given array on one line.

```
public static void main(String[] args) {
    int[] a1 = { 3, 1, 4 };
    int[] a2 = { 2, 0, 2, 5 };

    printArray(a1);    // Prints 3 1 4
    System.out.println();

    printArray(a2);    // Prints 2 0 2 5
}
```

2. Write a program named **WarmUp_11** that includes a method named **printReverse**. This method should print strings in a given array in reverse order on *one line* without any spaces separating them. Copy the following main method to your program:

```
public static void main(String[] args) {
    String[] a1 = {"t", "r", "a", "m", "s"};
    String[] a2 = {"r", "a", "t"};

    printReverse(a1);      // Prints smart
    System.out.println();

    printReverse(a2);      // Prints tar
}
```

👉 Please keep pressing **Ctrl+Shift+F**

3. Write a program named **WarmUp_12** that includes a method named **average**. This method should compute and return the average of values in a given array of integers. The average value should be a **double**. Copy the following main method to your program:

```
public static void main(String[] args) {
    int[] a1 = { 3, 7, 2 };
    int[] a2 = { 2, 0, 2, 5 };
    int[] a3 = { 5 };

    System.out.println(average(a1));    // Prints 4.0
    System.out.println(average(a2));    // Prints 2.25
    System.out.println(average(a3));    // Prints 5.0
}
```

4. Write a program named **WarmUp_13** that includes a method named **findLongest**. This method should find and return the longest string in a given array of strings. If there are more than one longest string, the method should return the first one. Copy the following main method to your program:

```
public static void main(String[] args) {
    String[] a1 = {"one", "hi", "hello", "ho", "house"};
    String[] a2 = {"goodbye", "hi", "how"};

    System.out.println(findLongest(a1));      // prints hello
    System.out.println(findLongest(a2));      // prints goodbye
}
```

Hint: You can apply the same generic algorithm that we use to find the minimum or maximum value in an array. Please refer your own notes from last week if you need recap.